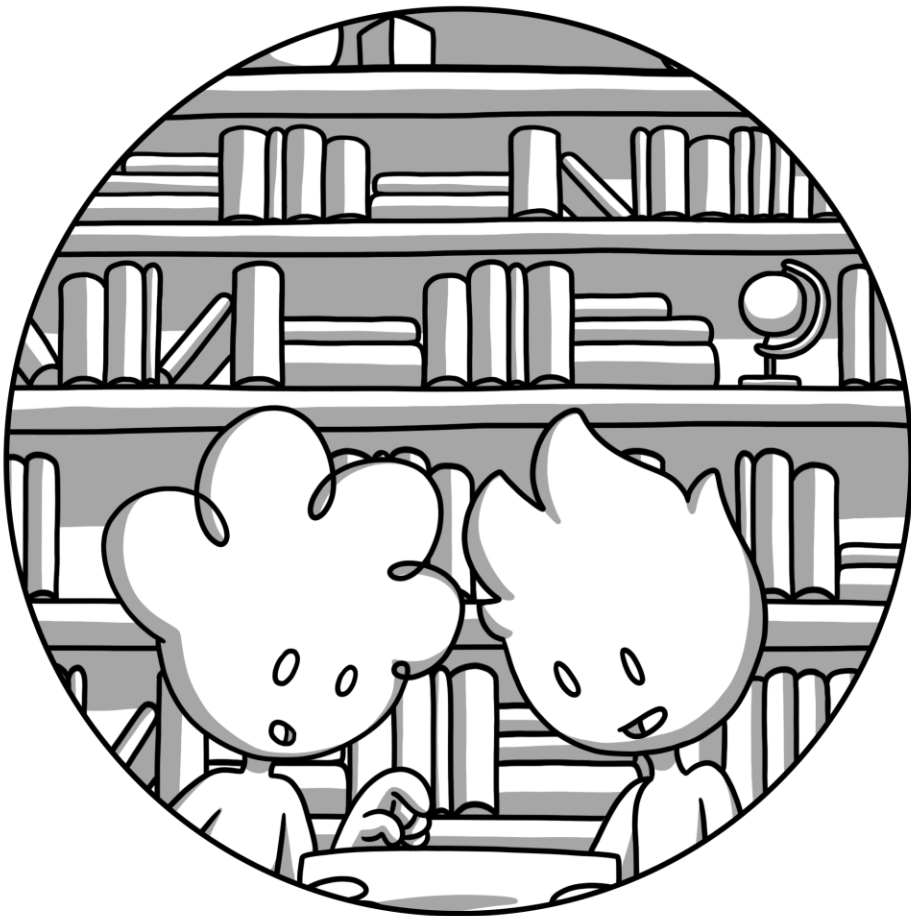


Attribute Types



COMIXPLAIN

This comic was created as part of the Comixplain research project, funded by the University of Applied Sciences St. Pölten (USTP) through the Innovation Call 2022.

Project Team:

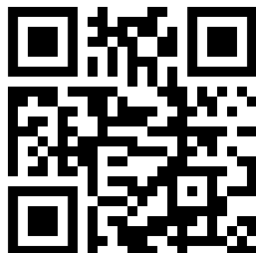
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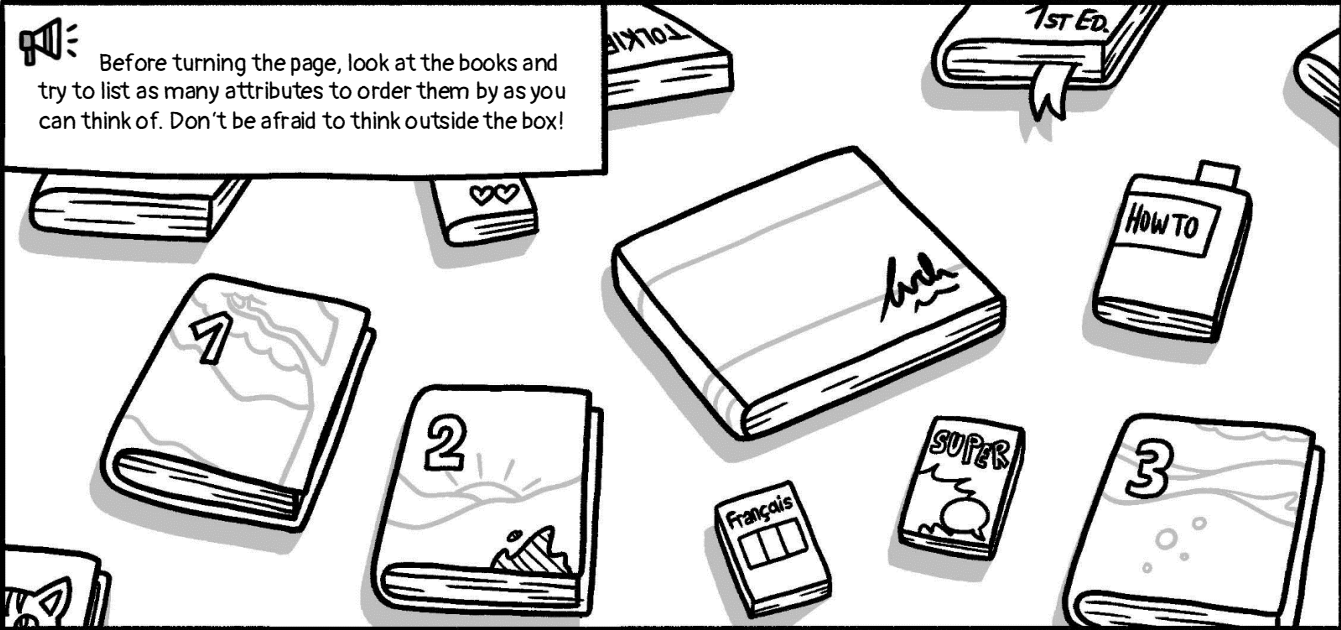
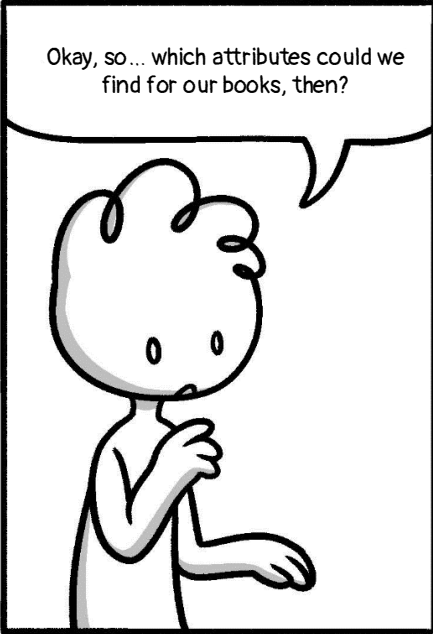
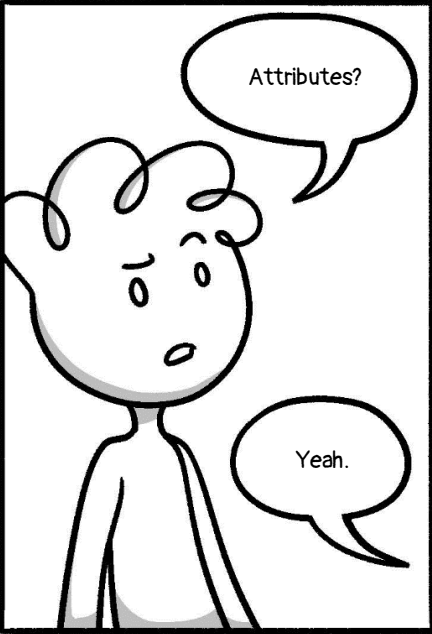
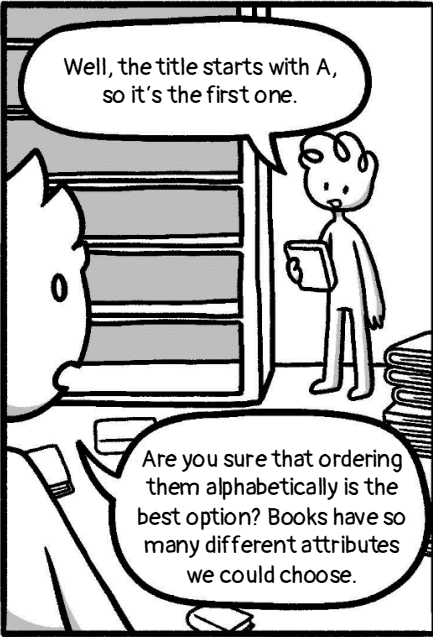
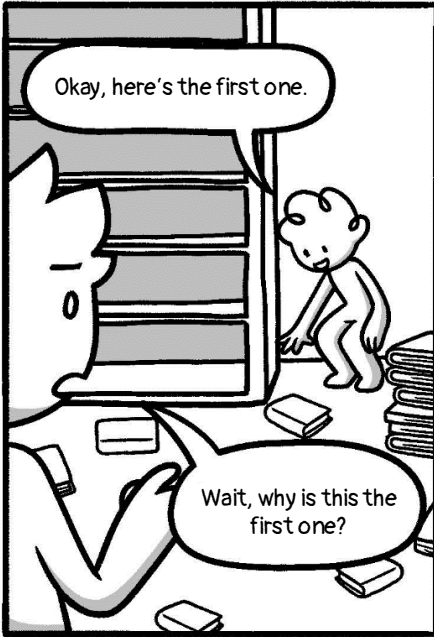
Magdalena Boucher & Alena Boucher



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(Victor-Adriel De-Jesus-Oliveira, Hsiang-Yun Wu,
Christina Stoiber, Magdalena Boucher, and Alena
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color, size, authors, material, part of series, year, pages, language, soft-/hardcover, genre, weight, edition, sign

Wow, that's so many attributes... what about color, then?

Sure! We could sort the books like the colors of the rainbow.

Or we could do it alphabetically by color name

It shouldn't be so hard to come to an agreement...

Yeah. Why is it that even if we agree on color, we can still order them differently?

I guess it's because of the attribute type...

Color hue is a **CATEGORICAL** attribute. It doesn't have an implicit order – just like language, genre, or author. We can order elements by categories, but among those, there is no given order.

Ah, I get it. So, what about the size of the books? That is definitely something with a sort of innate order, is it?

Exactly. Attributes can be categorical or ordered – and ordered attributes can be **ORDINAL** or **QUANTITATIVE**...

If you measure the books in centimeters, the measurement is **QUANTITATIVE**, as it supports arithmetic comparison.

20,5 cm is 8,4 cm smaller than 28,9 cm

The same goes for the weight in grams, or the number of pages.

But if I just split the books into groups of small, medium, or large...

... then the size is an **ORDINAL** attribute. We can't use it to calculate, but there is still a well-defined order.

small

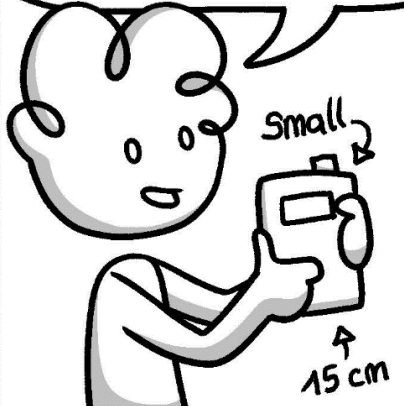
medium

large

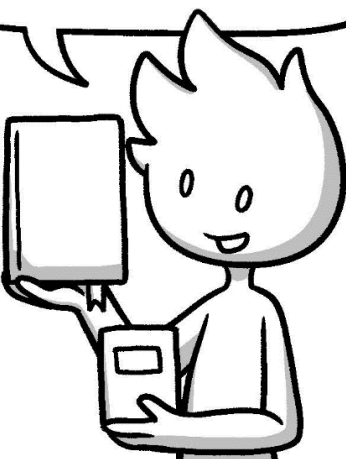
A variable can even be transformed to have another attribute type after you have collected the data.



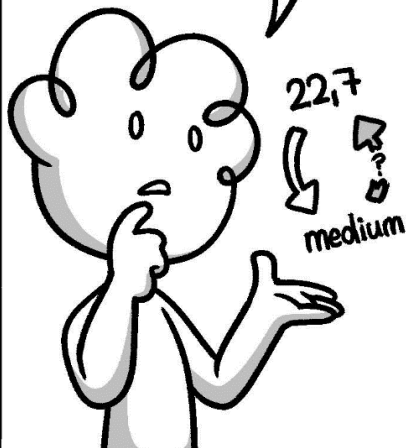
Like measuring all books in **centimeters**, and then deciding that every book under 18 centimeters is "**small**"? That would be transforming a **quantitative** attribute into an **ordinal** one.



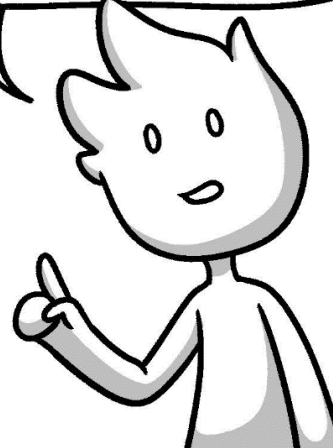
You're right. You could go even further and transform it into a binary categorical attribute: Pocketbooks or normal-sized.



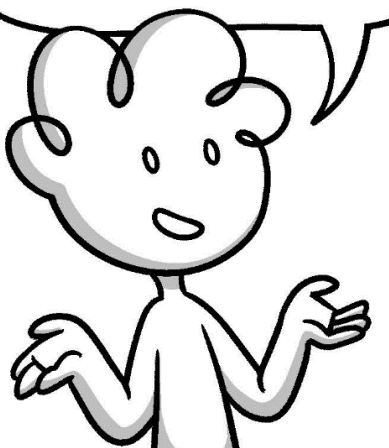
Oh, true. But doesn't that mean we'd lose most of the details of the data?



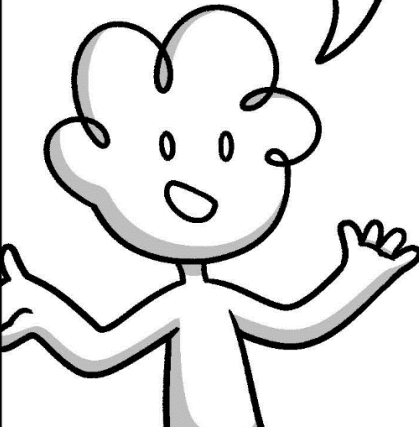
Yes, these transformations are *lossy*. Therefore, if you're not sure how you'll use the data, it might be better to collect it in more detail.



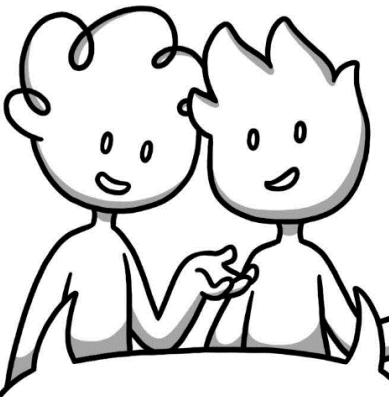
That makes sense. You can always transform a quantitative value, but if you only have categorical values from the start, you cannot switch to quantitative ones later.



Great, now that we both know everything about data attributes and attribute types, we can make an informed decision!



Sure, I will **attribute** that task to you and go grab a snack. Most of the books are yours, anyway.



Now that we have talked about data attributes and their types, turn the page and see if you can apply the knowledge in some exercises!

Remember: An attribute is a specific property that can be measured, observed, or logged. This graphic gives you an overview of the different attribute types.

ATTRIBUTES

CATEGORICAL

e.g., color, genre, author, language...

ORDERED

ORDINAL

educational level, shoe size, age group...

QUANTITATIVE

centimeters, kilograms, exact age, temperature...

Sources:
- Munzner, T. (2014). Visualization analysis and design. CRC press.
- Lazar, J., Feng, J. H., & Hochheiser, H. (2017). Research methods in human-computer interaction. Morgan Kaufmann.

Here is another exercise on attribute types: See how many different attributes you can find for this assortment of chocolates!

PRALINE

E.g.: size, weight, shape, color, flavor, calories, sugar/fat..., number of nuts, type of wrapping/filling/chocolate..., expiration date, consistency, (not) handmade...

Most of the time, data is presented to you in tables. Try to guess the attribute types of each row!

Title	Director	Actor	Length	Year	Popularity	Genre
Goldfinger	Hamilton	Connery	112	1964	7.7	Action
Ben Hur	Wyler	Heston	212	1959	8.2	Action
Ben Hur	Niblo	Novarro	133	1926	7.4	Drama
Gladiator	Scott	Crowe	155	2000	8.5	History
Casablanca	Curtiz	Bogart	102	1942	9.5	Romance

Categorical: Title, Director, Actor, Genre | Quantitative: Year, Length, Popularity

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Sources:

Munzner, T. (2014). Visualization analysis and design. CRC press.

Lazar, J., Feng, J. H., & Hochheiser, H. (2017). Research methods in human-computer interaction. Morgan Kaufmann.

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